1. Stock Maintenance System

1.1 Introduction

Managing inventory in a growing business is a delicate balancing act. Hold too little stock, and you risk stock-outs, delayed orders, and unhappy customers. Hold too much stock, and valuable capital gets tied up in warehouse space, carrying costs, and expired or obsolete goods.

Relying on manual spreadsheets or physical logbooks to track stock movements quickly becomes a recipe for inventory discrepancies, misplaced items, and fulfillment bottlenecks. A modern **Stock Maintenance System (SMS)** replaces manual guesswork with real-time operational control. By automating item cataloging, purchase orders, goods receipts, warehouse transfers, and low-stock alerts, the platform gives storekeepers, procurement teams, and managers complete visibility into what is in stock, where it is stored, and when it needs to be reordered.

1.2 Software Requirements Specification (SRS)

1.2.1 Purpose of this Document

This SRS document outlines the functional, technical, and operational requirements for developing the Stock Maintenance System. It acts as a single blueprint for software engineers, database architects, warehouse managers, purchasing staff, and business executives, ensuring complete alignment on system capabilities, constraints, and delivery goals.

1.2.2 Scope of the System

The system governs the full lifecycle of warehouse inventory across multiple locations. This includes SKU master management, vendor directory tracking, purchase order generation, goods receiving, stock issue logging, inter-warehouse transfers, damaged/expired stock write-offs, reorder point automation, and stock valuation reporting.

- **Estimated Project Cost:** $40,000
- **Target Delivery Timeline:** 6 Months

1.2.3 System Overview

The platform serves as an integrated inventory workspace connecting warehouse personnel, procurement specialists, sales teams, and accounting systems. By automating routine inventory tracking and ledger updates, it prevents manual entry errors and minimizes carrying costs. Key modules include:

- **Item Master & Catalog Management**
- **Supplier Directory & Procurement Engine**
- **Inward Goods Receipt & Quality Check**
- **Stock Issue & Fulfillment Control**
- **Warehouse & Location Management**
- **Automated Reorder & Threshold Alerts**
- **Inventory Valuation & Financial Audit Ledger**

1.2.4 General Description

Core Functions

- **SKU & Catalog Management:** Maintain master records for all items, including categories, unit pricing, reorder thresholds, safety stock levels, and bin/rack locations.
- **Procurement & Goods Receiving:** Raise purchase orders (POs) to approved vendors, record incoming shipments via Goods Received Notes (GRN), and automatically update stock quantities upon inspection.
- **Dispatch & Stock Issue:** Log stock issued for customer orders or internal department use while enforcing real-time availability checks to prevent negative inventory.
- **Transfers & Adjustments:** Record stock transfers between warehouse locations, log damage or expiry write-offs with mandatory reason codes, and maintain an immutable audit trail.
- **Automated Reorder & Analytics:** Continuously monitor stock levels to generate reorder alerts, highlight slow-moving or expiring inventory, and produce accurate valuation reports.

User Objectives

- **Warehouse & Store Staff:** Speed up goods receiving and order picking using handheld barcode/RFID scanners, ensuring accurate bin placement and real-time stock updates.
- **Procurement Staff:** Automate purchase order generation, monitor vendor lead times, and replenish stock before inventory reaches critical levels.
- **Business Managers & Executives:** Gain real-time visibility into overall stock value, turnover rates, shrinkage metrics, and holding costs to make smarter buying decisions.

System Users

- **Storekeepers & Warehouse Operators:** Staff responsible for receiving, scanning, storing, picking, and dispatching inventory items.
- **Purchase Officers & Buyers:** Staff who negotiate with suppliers, issue purchase orders, and monitor delivery schedules.
- **Warehouse & Finance Managers:** Supervisors who approve purchase orders, review stock adjustments, and analyze financial inventory valuation reports.
- **System Administrators:** IT staff managing user accounts, security roles, barcode scanner hardware configurations, and system backups.

Key Benefits

- Elimination of stock-outs and overstocking through automated safety-stock reorder alerts.
- Faster order processing and receiving using barcode and mobile scanning integration.
- Drastic reduction in inventory shrinkage, loss, and unrecorded damages.
- Accurate financial inventory valuation (FIFO / Weighted Average) for seamless accounting integration.

1.2.5 Functional Requirements

- **Item Master & Category Management**

    - Maintain unique Stock Keeping Units (SKU) with attributes: Item Name, Category, Unit of Measure (UOM), Cost Price, Selling Price, Safety Stock, Reorder Point, and Default Bin Location.
    - Support batch/lot tracking and expiration date monitoring for perishable or regulated goods.
    - Maintain real-time stock balances across individual warehouse bins and overall facility locations.
- **Supplier & Procurement Management**
    - Maintain vendor profiles, contact details, lead times, payment terms, and performance history.
    - Create, approve, and issue Purchase Orders (PO) with automated tax, discount, and total cost calculations.
    - Track PO status lifecycle (**Draft, Issued, Partially Received, Fully Received, Closed, Cancelled**).
- **Stock Receipt (Inward) & Quality Inspection**
    - Process Goods Received Notes (GRN) matched against open Purchase Orders.
    - Support partial receipts, log damaged goods during shipment, and automatically update available stock levels upon receipt approval.
- **Stock Issue (Outward) & Fulfillment**
    - Record stock dispatches for sales orders, internal consumption, or inter-departmental transfers.
    - Enforce strict availability checks to block stock issue attempts exceeding available on-hand quantities.
- **Warehouse Transfers & Stock Adjustments**
    - Support inter-warehouse stock transfer workflows with transit tracking and destination receiving confirmation.
    - Log manual stock adjustments (damaged, lost, expired, or audited inventory) with mandatory reason codes and supervisor sign-offs.
- **Automated Reorder Alerts & Inventory Intelligence**
    - Continuously compare on-hand and on-order quantities against defined reorder points.
    - Trigger automated system notifications (email/SMS/dashboard alerts) and generate pre-populated draft Purchase Orders for low-stock SKUs.
    - Flag slow-moving inventory, dead stock, and approaching product expirations.
- **Valuation & Reporting Engine**
    - Calculate dynamic inventory valuation using FIFO (First-In, First-Out) or Weighted Average Cost methods.
    - Generate scheduled and on-demand reports covering: Current Stock Summary, Stock Movement Ledger, Reorder Analysis, Supplier Performance, Expiry Risk, and Shrinkage Audits.

1.2.6 Interface Requirements

- **Software Interfaces**
    - **Accounting & ERP Integration:** Secure APIs to export purchase ledgers, stock valuation journals, and accounts payable entries to enterprise accounting systems.

    - **Barcode / RFID Peripheral Drivers:** Integration with mobile handheld devices, barcode scanners, and thermal barcode label printers.
    - **Notification Gateways:** SMS and email notification APIs for sending automated reorder alerts and purchase order copies to vendors.
    - **Business Intelligence & Reporting Tools:** Data export capabilities (PDF, CSV, Excel) for custom analytics.
- **User Interfaces**
    - **Desktop Operations Dashboard:** High-density, keyboard-shortcut-friendly interface for purchasing officers, store managers, and system admins.
    - **Mobile Handheld Interface:** Lightweight, mobile-optimized UI designed for wireless handheld barcode scanners used during warehouse picking, put-away, and stock counts.

1.2.7 Performance Requirements

- **User Capacity:** Support up to **300 active concurrent user sessions** without latency issues.
- **Real-time Synchronization:** Update physical and digital stock levels within **2 seconds** of a barcode scan or transaction commit.
- **System Uptime:** Maintain **99.5% operational uptime**.
- **Processing Accuracy:** Maintain a transaction processing error rate below **0.05%**.

1.2.8 Technical & Design Constraints

- **Hardware & Infrastructure Constraints**
    - Dedicated server infrastructure with a minimum of 16 GB RAM, 1 TB SSD storage, and an automated off-site backup system.
    - Hardware compatibility with standard wireless handheld barcode/RFID scanners and industrial label printers.
- **Software Constraints**
    - Developed using robust, secure frameworks (**Java / Spring Boot** or **Python / Django**).
    - Relational database engine using **PostgreSQL** or **MySQL** with strict ACID transaction guarantees for inventory ledgers.
- **Regulatory & Accounting Constraints**
    - Compliance with standard commercial tax, GST/VAT invoicing, and financial audit guidelines.
    - Adherence to local data protection laws regarding vendor and customer profile storage.

1.2.9 Non-Functional Quality Attributes

- **Security:** Role-Based Access Control (RBAC), encrypted transmission (TLS 1.3), and immutable audit logging for every inventory adjustment or quantity update.
- **Portability & Cross-Platform Support:** Web interface compatible with major desktop and mobile web browsers across Windows, Linux, Android, and iOS.
- **Reliability & Resilience:** Daily automated database backups with quick disaster recovery protocols to prevent data corruption.

- **Scalability:** Modular database architecture designed to support new warehouse locations, additional SKUs, and higher transaction throughput as business scales.
- **Data Integrity:** Strict relational foreign-key constraints to guarantee inventory ledgers match physical receiving and dispatch transactions without discrepancies.

1.2.10 Project Schedule and Budget

Project Timeline (6 Months Total)

1. **Requirements Gathering & Inventory Process Mapping:** 1 Month
2. **System Architecture, Database Schema & UI/UX Design:** 1 Month
3. **Core Application Development & Hardware Integration:** 3 Months
4. **Testing, QA, Warehouse Staff Training & Deployment:** 1 Month

Budget Allocation

- **Core Application Development:** $24,000
- **Quality Assurance, Hardware Integration & Testing:** $8,000
- **Deployment, On-Site Setup & Initial Maintenance:** $8,000
- **Total Estimated Budget: $40,000**
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1.3 Class Diagram:

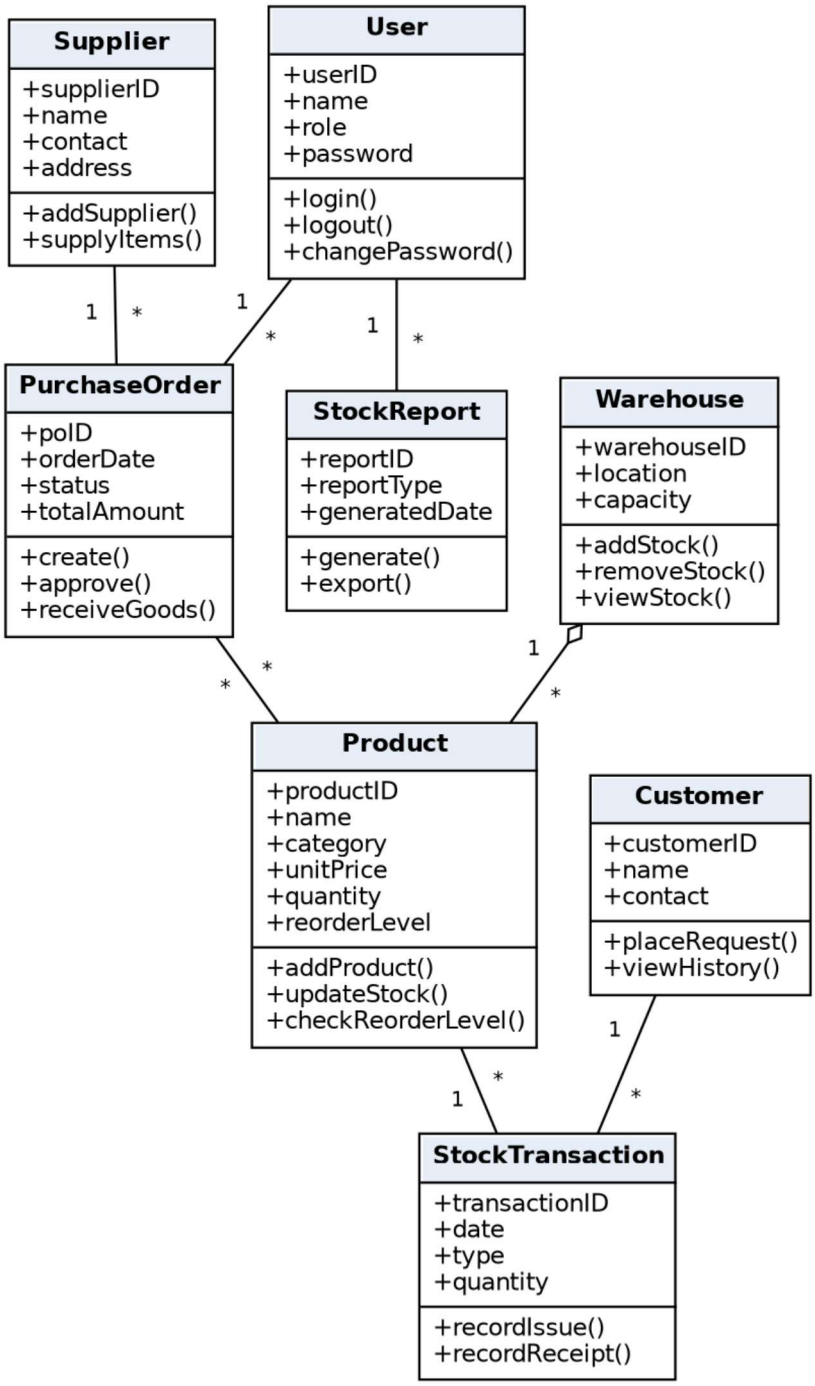


Fig 1.1

Brief Description:

The class diagram for the Stock Maintenance System (SMS) represents the key entities, their attributes, and relationships in the system. Below are the main components typically included in the class diagram for SMS:

- **User:** represents system users (store staff, purchase staff, managers) with role-based login.

- **Product:** holds item details, quantity on hand and reorder level.
- **Supplier:** stores supplier details and supplies products against purchase orders.
- **PurchaseOrder:** records items ordered from a supplier with date, status and total amount.
- **Warehouse:** represents a storage location that holds products.
- **Customer:** represents the party to whom stock is issued or sold.
- **StockTransaction:** records every receipt or issue of a product with date and quantity.
- **StockReport:** generated by users to analyse stock levels, movement and valuation.